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MILITARY SPECIFICATION

TELESCOPE, MARK 100 MOD 0 AND 1

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers the construction and performance requirements for a prismatic binocular telescope intended for use aboard ship in Gun Director Mk 68 for sighting and tracking both aerial and surface targets.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in me *issue of* the Department of Defense of Specifications and Standards (DoDISS) and supplements thereto cited in the solicitation.

SPECIFICATIONS

FEDERAL

HH-P-96	Paper, Gasket Fiber (Animal or Plant), Sheet
QQ-C-390	Copper Alloy Castings (Including Cast Bar)

Military

MIL-P-116	Preservation, Methods of
MIL-T-152	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-C-675	Coating of Glass Optical Elements (Antireflection)
MIL-S-901	Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, Maryland 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1240

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MIL-E-2036	Enclosures for Electric and Electronic Equipment, Naval Shipboard
MIL-P-3584	Plastic Sheet, Polyvinyl Chloride Plasticized, Elastomeric
MIL-P-5516	Packing, Preformed, Petroleum Hydraulic Fluid Resistant, 160° F
MIL-C-6183	Cork and Rubber Composition Sheet for Aromatic Fuel and Oil Resistant Gaskets
MIL-F-7179	Finishes, Coatings, and Sealants for the Protection of Aerospace Weapons Systems
MIL-O-13830	Optical Components for Fire Control Instruments General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-E-15090	Enamel, Equipment, Light Gray (Formula NO. 111)
MIL-E-16400	Electronic Interior Communication and Navigation Equipment, Naval Ship and Shore General Specification for
MIL-F-18870	Fire Control Equipment, Naval Shipboard, General Specification for
MIL-F-21424	Filter, Polarizing (for Optical Instruments)
MIL-G-23827	Grease, Aircraft and Instrument, Gear and Actuator Screw, NATO Code Number G-354, Metric

STANDARDS

FEDERAL

FED-STD-H28	Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-109	Quality Assurance, Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167-1	Mechanical Vibrations of Shipboard Equipment (Type 1- Environmental and Type II - Internally Excited)
MIL-STD-210	Climatic Information to Determine Design and Test Requirements for Military Systems and Equipment -
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-1303	Painting of Naval Ordnance Equipment

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2.1.2 Other Government drawings and publications The following other Government drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND

8-Z-1008 to 8-Z-1021	Screws for optical Instruments
8-Z-1157	Air Valve
LD167920	List of Drawings for Telescope Mark 100 Mod 0
LD258406	List of Drawings for Telescope Container (Storage Box)
LD412528	List of Drawings for Telescope Mark 100 Mod 1
510543	Headrest Mount
510544	Headrest Support
511050	Headrest

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND

OP 1959	Telescope Mark 100 Mod 0, Description and Instructions for Installation, Alignment, Operation, Maintenance, and Overhaul
OP 2230	Workmanship and Design practices for Electronic Equipment
OP 3750	Optical Shop Practice

(Copies of specifications, standards, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article When specified in the contract or purchase order a sample shall be subjected to first article inspection (see 4.4 and 63).

3.2 Construction. Telescope Mk 100 Mods 0 and 1, which hereinafter may be referred to as simply the telescope, shall be constructed as specified herein and in accordance with the drawings and specifications listed on LD167920 for Telescope Mk 100 Mod 0 and on LD412528 for Telescope Mk 100 Mod 1.

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3.2.1 Mechanical requirement The telescope shall be constructed to have the eyepiece in a horizontal plans parallel to the plane formed by the horizontal mounting fast on the bottom of the instrument. The line of sight shall be capable of being elevated 95 defrees above this plane and depressed 25 degrees below the same plane. Movement of parts involved in the operation of the telescope shall be positive,accurate, free from excessive friction, and shall not transmit shock of jar to optical elements of their monts. Operaating knobs and levers shall be capable of being adjusted by gloved hands.The mirror and its assembly shall be so balanced as to maintain and hold its position for any setting between the limits of elevation specified for environmental conditions set forth herein. The torque required to turn the elevation mirror adjusting shaft shall not exceed 5 inch-pounds

3.2.1.1 Limit stops Limit stops shall be provided to prevent derangement of mechanisms. The travel of the line of sight of the telescope shall be limitedby such stops, one stop being set at + 100 degrees (elevation) and the other -30 degrees (depression).

3.2.1.2 Alignment of elements Means shall be provided for checking, aligning, and adjusting all interrelated elements of the telescope.

3.2.1.3 Headrest and support Each telescope shall be fitted with a headrest support similar in principle to that shown on Drawings 510543 and 510544. On this support shall be mounted the soft rubber headrest shown on Drawing 511050 (see 37).

3.2.1.3.1 Eye buffer. Each ocular shall have provisions for mounting a soft rubber eye buffer or eye guard as specified in MIL-F-18870. Eye buffers or wyw guards shall be supplied by the contractor.

3.2.1.4 Metallic parts and materials. The main body casting shall be of copper alloy No. C92300, in accordance with QQ-C-390. Other metal parts shall be of brass or bronze, in conformance with specifications listed on LD167920 or LD412526 as applicable unless permission is obtained from the conrtacting activity for the use of other material which may be satisfactorily substituted. The selection of dissimilar metals for use in construction of the telescope shall be made to minimize corrosive action caused by electrolyte action between the metals forming a galvanic couple. Possible galvanic couples and the degree of corrosive action associated with each are listed in MIL-E-16-400

3.21.4.1 Non-metallic parts and materials. Non-metallic parts and material used in the construction of the telescope shall be in accordance with the drawings and specifications listed on LD167920 or LD412528 as applicable and as specified herein. Materials which are nutrients for fungus shall not be used in applications other than those approved in MIL-STD-454 unless prior approval of the contracting activity has been obtained. Nutrient materials, when used, shall be treated in accordance with MIL-T-152

3.21.5 Lens mountings. All lenses shall be secured by means of screwed-in retaining rings, with lock rings or set screws. "Spinning of beating metal over the edges of lenses will not be acceptable. The objective lens shall be provided with a double eccentric mount for collimation.

3.2.1.6 Screws. All screws shall conform either to FED-STD-H28 or to those shown on Drawings 8-Z-1008 to 8-z-1021. All threaded parts shall conform to FED-STD-H28 as regards pitch, form, and size of thread.

3.2.1.7 Finish. All external metal surfaces, except for mounting surfaces, shall be covered with two (2) of light gray enamel in conformance with MIL-E-1509Q Operating knobs handles, etc, shall be covered with a durable black baked lacquer, system no. 9, in accordance with MIL-SD-1303

3.2.1.8 Reference designations and markings. All parts of the telescope shall be marked as shown on applicable drawings listed in LD167920 or LD412528 as applicable and as specified herein. Markings shall include an identification plate affixed on the surface of the central portion of the telescope In lieu of the identification plate such data as would be contained therein maybe engraved on the exterior of the body casting provided precautions are taken to proect the engraved figures and numbers from being defaced by corrosion or subsequent painting of the telescope. Identification will not be required for glass components.

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3.2.1.9 Degree of enclosure. The enclosure for the optical components of this telescope shall be airtight as defined in MIL-E-2036

3.2.1.9.1 Air inlet valve and outlet plug. The body of the telescope shall be provided with an air inlet valve in accordance with Drawing 6-2-1157 and an outlet plug, the two located so that the interior of the telescope may be dried out by a current of dry air or other gas. The air inlet valve and the outlet plug shall be located at the eyepiece end of telescope for easy access during regassing operation Suitable vents and bypasses shall be provided in the interior of the body so that the air or gas has access to all optical surfaces The words “air outlet” shall be engraved near the outlet plug.

3.2.1.9.2 Sealing and gaskets.

3.2.1.9.2.1 Sealing large windows. Material used for sealing under large windows shall be in accordance with HH-P-96

3.2.1.9.2.2 Large window gaskets Material used for gaskets between large windows and their retainers and under all cover plates shall be 0.045 inch thick and shall be in accordance with MIL-C-6183 The width of the window ledge and also that potion of the window retainer in contact with the glass shall be at least as wide as the thickness of the glass window used.

3.2.1.9.2.3 Charging valve gaskets. Material used for gaskets at charging valves shall be 1/16 inch thick and shall be in accordance with MIL-P-3584.

3.2.1.9.2.4 Sealing screws. At headless setscrews and for washers under the heads of external screws such as three for retaining large windows, "Prestite” wax, or equal, shall be used.

3.2.1.9.2.5 Illuminator window gaskets Material for gaskets under illuminator windows shall be 1/32 inch thick and shall be in accordance with MIL-P-3584.

3.2.1.9.2.6 Sealing operating shafts. For Telescope Mk 100 Mod 0 the packings at polarizing filter operating shafts where they enter the airtight sections of the telescope shall be sealed by using “Hycar 1232-HGR-2" or equal.

3.21.9.2.7 O-ring gaskets For Telescope Mk 100 Mod 1, the use of O-ring gaskets will be permissible provided the gaskets conform to MIL-P-5516 and are of the standard sizes listed and shown thereon.

3.2.1.10 Lubrication. Provisions shall be made to lubricate all bearing and wearing surfaces Lubricants to be used shall be selected in accordance with MIL-G-23827.

3.3 Optical requirements.

3.3.1 Optical characteristics The optical characteristics of the telescope shall be as follows:

Dual magnification.	10 diameters ± 5% 6 diameters ± 5%
True field	for 10x, approximatety 7°10’ for 6x, approximately 12°’5’
Exit pupil	for 10x, approximately 6 mm for 6x, approximatety 7 mm
Eye distance	25 to 27 mm at either magnification

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3.3.2 Image quality and definition. When subjected to a star test as specified herein, the image shall appear round with a sharply defined edge The image shall be free from points flare, rays, fog, haze or halos other than the normal diffraction rings. No difference in coloration between one side of the star and the other shall be seen A slightly defocused image" shall remain round The ocular setting between the three stars shall not exceed 0.2 diopter.

3.3.3 Resolution. The limit of resolution at any point within 1 degree of true field from the field center shall be 6 seconds or batter for a magnification of 10 diameters and 10 seconds or better for a magnification of 6 diameters for parallel lines of any orientation.

3.3.4 Reticle A reticle shall be provided for use at either magnification and shall be mounted in the right optical system of the telescope The reticle pattern at either magnification shall consist of two centrally located concentric circles. The dimensions of the reticle pattern will be specified by the contracting activity (see 6.2.1). The lines shall be clean-cut, unbroken, and filled with suitable hard, durable and opaque material. The rightoptical system containing the reticle shall be free from parallax on a distant target.There shall be a nameplate, mounted in a conspicuous location on the telescope. which will give the diameter of each of the circles of the reticle pattern in true field at 6 and also at 10 diameters magnification.

3.3.4.1 Reticle illumination. Artificial illumination shall be provided for the reticle Such illumination shall be bymeans of a Mazda,64" 6 to 8 volt 3 candle power lamp with contact base and spherical bulb 3/4 inch in diameter. A Lamp Socket Mark 9, as listed on LD16792Q for this lamp will be provided by the contracting activity, but the contractor shal supply the holder for this socket Illumination shall provide substantially uniform illumination of both circles of the reticle at either magnification. Furtermore, no bright and troublesome reflections shall be visible in the field of view nor at a distance in the dark.

3.35 Reflection reducing films- The air-toglass surfaces of each element of the optical system of the telescope, excluding both surfaces of teh reticle plate, or glass surfaces lying in a foclal plane,shall be treated with a reflection-redudng film in accordance with MIL-C-675.

3.3.6 Stray light. No stray or scattered light, or reflections from surfaces within the telescope shall be visible in the field of view, or enter the exit pupil, or be visible in the vicinity of the exit pupil when the pupil is viewed from a distance of 2 0r 3 feet along the Ocular axis extended. All internal surfaces from whichstray or scattered light might be reflected shall be threaded, knurled, grooved or scored in such manner as to avoid such reflection. Such surfaces, and any other surfaces from which troublesome reflection might occur, shall be darkened by treatment. No paint, enamel, lacquer, or other material which might become detached from the surface to which applied, shall be used in the interior of these telescopes.

3.3.7 Collimation.

3.3.7.1 Parallelism. When two parallel pencils or bundles of light rays are projected into the two objective lenses respectively, the conjugate rays emerging from the two eyepieces shall be parallel within the following limits for any setting of the interocular distance scale:

a. Fourteen minutes of arc (equivalent to 1.4 minutes of true field at 10x and 2.3 minutes at 6x) in the direction perpendicular to the plane determined by the two eyepiece axes.

b. Twenty-eight minutes of arc (equivalent to 2.8 minutes of true field at 10x and 4.6 minutes at 6x) divergence and fourteen minutes of arc (equivalent to 1.4 minutes of true fieid at 10x and 2.3 minutes at 6x) convergence in a direction parallel to the plane of the two eyepiece axes.

3.3.7.2 Squaring. The images of a straight tine formed by the two optical systems, at either magnification, shall be parallel within one-half degree.

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3.3.8 Eyepiece focus. Each eyepiece of this telescope shall be capable of being focused from plus 1 to minus 2 diopters by rotation of a ring or knob which moves an internal lens component. A focusing scale and index shall be provided. Each line of the focusing scale shall be correctly located within 1/4 diopter and each interval correct within 1/4 diopter. The eyepieces shall be airtight.

3.3.9 Filters Each optical system of the telescope shall be provided with a variable density polarizing filter, so arranged that when the planes of polarization of the two plates in each filter are parallel, this common plane shall be so oriented as to reduce surface glare in the target field to a minimum. A single-knob control shall be provided which simultaneously operates the filters in each optical system of the telescope. The polarizing filter shall comply with MIL-F-1424.

3.3.10 Interocular adjustment. Provision shall be made for manual adjustment of the intercular distance from 58 to 72 millimeters by movement of a control lever or knob. A scale graduated in 2 millimeter internals shall be Provided. The interpupillary distance indicated by this scale shall be correct to within 1/2 millmeter.

3.3.11 Accuracy. The accuracy requirements with respect to elevation and depression of the line of eight shall be as follows:

a. The plane established by the line of sight from -25 degrees to +95 degrees shall be perpendicular to the plane formed by the four horizontal machined surfaces on the feet of the telescope body casting within one minute of arc.

b. The plane established by the line of sight from -25 degrees to +95 degrees shall be parallel to the plane formed by the two vertical machined surfaces on the left feet of the telescope body casting within one minute of arc.

c At ny elevation or depression, the elevation of the line of sight with respect to plane formed by four horizontal machined surfaces or the feet of the telescope body casting shall correspond to the amount of rotation of the input shaft within two minutes of arc.

d. The above requirements shall be met at either magnification, with the variable density filters set for any value of transmission in the telescope field of view and also with these filters removed, i.e., clear.

Machining of the mounting surfaces of a telescope after assembly of the optical parts therein, in order to secure proper alignment, will not be permitted.

3.4 Environmental requirements.

3.4.1 Shockproofness. The telescope shall exhibit the degree of shockproofness required for the appropriate grade classification, and class of equipment specified in MIL-S-901 as follows:

Degree of Shockproofness	Grade A
Equipment classification	Subsidiary component
Equipment class	Class I
Test classification	Lightweight
Type of test	Type B

3.4.2 Vibration. The telescope shall be capable of withstanding Type I environmental vibration as set forth in MIL-STD-167.

3.4.3 Temperature and humidity. The telescope shall be capable of withstanding without damage or impairment of performance temperature and humidity extremes as specified in MIL-STD-210 for worldwide shipboard operation or as specified for Class 2 equipment in MIL-F-18870. The telescope shall also withstand temperature and humidity conditions of short-term storage and transit as set forth in MIL-STD-210

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3.4.4 Static loading. The telescope shall be capable of withstanding static loading of 1,000 pounds per square inch in any and all directions.

3.5 Government furnished equipment if specified in the contractor order (see 6.2.1), certain items such as headrests and lamp sockets may be supplied as Government furnished equipment.

3.6 Government loaned equipment Government equipment and property will be provided as indicated in the contract or order (see 6.2.1) and normally will include only peculiar end items not available commercially for factory test and evaluation of the telescope.

3.7 Workmanship All workmanship shall be in accordance with OP 2230 and as specified herein. The standards of workmanship exhibit in any first article sample subject to any qualification stated in the Government's notice of approval shall be determinative of the requirements of the contract relative to workmanship insofar as not specificity covered by applicable specifications.

4. QUALITY ASSURANCE PREVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.21), the contractor is responsible for the performance of ail inspection requirements as specified herein. Exdcept as otherwise specified in the contractor purchase order (see 6.2.1), the contract or may use his own or any ther facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of me inspections set forth in the specification where such inspections are deemed necessary to assure suppiles and services conform to prescribed requirements.

4.1.1 Responsibility for compliance All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that ail products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 45).

4.3 Inspection conditions. Unless otherwise specified, ail inspections shall be performed in accordance with the test conditions specified in MIL-STD-109.

4.4 First article inspection. Unless Otherwise specified in the contract or order (see 6.2.1), a first article sample shall be subjected to the quality conformance inspections of 45345.4, and 455

4.4.1 Sampling. The first article sample shall be a telescope manufactured in accordance with the requirements of section 3 of this specification.

4.4.2 Rejection. Failure of the sample to meet all the requirements of this specification shall be cause for rejection of the first article sample

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4.5 Quality conformance. Except as otherwise specified herein, each telescope shall be subjected to examinations and tests specified herein to assure compliance with the requirements of this specifcation.

4.5.1 Sampling. Each telescope offered for acceptance under contract Shall be subjected to the examinations and tests specified in 45.3 and 4.5.4. In addition, samples selected from units having satisfactorily passed the inspections of 4.5.2 and 4.5.4 shall be subjcted to the teats of 4.5.5 at an activity designated by the contracting officer. Sampling for the tests of 455 shall be as determined by the contract or order (see 6.2.1).

4.5.2 Rejection. Failure to meet any of the specified Inspection requirements shall be considered cause for rejection of the telescope.

4.5.3 Examintions.

4.5.3.1 Visual and dimensional examination. Each telescope shall be visually and dimensionally examined for completeness of manufacture, freedom from defects, proper identification, and workmanship Dimensional and other non-destructive inspections shall be made to verify that the telescope and its container are in accordance with this specification and the drawings and specifications listed on LD167920 or LD412528 as applicable Examinations shall be conducted in accordance with the appicable portions of MIL-0-13830.

4.5.3.2 Internal cleanliness Examination for internal cleanliness shall be made by looking into the telescope in bright tight, both directly and with the aid of an auxiliary lens placed behind the eyepiece and by looking through the telescope at a uniform field having the brightness of the sky in average daylight.

4.5.3.3 Stray light. With the telescope directed at a distant natural target silhouetted against a bright sky at least 40 degrees from me direct line of the sun, the exit pupil shall be examinad with the aid of a 5-power loupe. Upon placing the eye in the exit pupil, no haze or clouding of the image resulting from superimposed stray light shall be evident as the observer's eye is slowly moved out of the exit -pupil in a lateral direction, until the image is extinguished.

4.5.3.4 Adjustments and other operational features. Operational features such as knobs, controls, and illumination shall be examined to determine conformance with this specification.

4.5.3.4.1 be of sight adjustmenta Line of sight adjustments shall be dimensionally verified to be 95 degrees and 25 degrees above and below, respectively, the horizontal plane of the telescope The limit stops shall also be dimensionally examined to ascertain that they are set at +100 degrees (elevation) and -30 degrees (depression).

4.5.3.4.2 Mirror assembly. The mirror and its assembly shall be operated through the limits of elevation. The torque required to turn the adjusting shaft shall not exceed 5 inch-pounds.

4.5.3.4.3 Reticle illumination. The reticle lamp shall be energized and the reticle shall be observed to escertain that substantially uniform illumination of both circles of the reticle at either magnification is effected. No bright and trousblesome reflections hall be visible field of view nor distance in the dark.

4.5.3.5 Controls. All operating controls shall be manipulated through their ranges to determine that movement of parts involved in the operation of the telescope is positive accurate free from excessive friction, and does not transmit shock or jar to optical elements During this examination observe that operation of controls can be pro ly conducted with gloved hands.

4.5.4 Performanoe tests. Performance teats specified herein are not set forthwith regard to any particular sequence The contractor shall establish appropriate sequence for conducting inspections specified herein to ensure that the requirements of this specification are met.

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4.5.4.1 Measurements during all tests, the accuracy of instruments and test equipment used to control or monitor test parameters shall be verified periodically to the satisfaction of the contracting activity. All instruments and test equipment used in conducting tests specified herein shall:

a. Conform to laboratory standards where calibration is traceable to the prime standards at the U.S Bureau of Standards

b. Have an accuracy of at least one-fifth the tolerance for the variable to be measured.

4.5.4.2 Standard conditions. All quality conformance tests shall be conducted under standard conditions of ambient temperature atomospheric pressure, and relative humidity. Standard conditions shall be as foiiows:

Temperature	25 ± 10 degreesc Celsius (c)
Relative humidity	90 percent or less
Barometric pressure	28 t0 32 inches of mercury
Telescope internal gas pressure	2 pounds per square inch

Actural ambient conditions shall be recorded periodically during th test period and provided as part of the test results.

4.5.4.3 Collimation. Collimation of the telescope shall be made as described in OP 1959. The procedures desribed therein are directed toward Collimation after overhaul of an in-service telescope but shall be utilized in verifying the following requirements of this specification

a. parallelism
b. squaring
c.accuracy
d. magnification
e. true fieid
f. exit pupil
g. eye distance
h. focus
i. resolution
j. reticle
k. interocuiar adjustment

Tests to verify the above requirements shall be in accordance with MIL-13830, where applicable Otherwise tests shall be as specified herein.

45.4.4 Image quality and definition. The telescope shall be subjected to a star test to determine that the image quality and definition is as specified herein.

4.5.4.4.1 Test equipment The target for the star test shall be situated at least 50 meters from the test site or a miniature target maybe contained in a suitable collimator. The target shall consist of three adjacent circular apertures, each 10 seconds in angular diameter, placed beside one another. All of the apertures shall be brightly illuminated from by means of tungsten or lamp operated at ordinary line voltage. The light from the first, or white star, shall be unfiltered. The light emitted by the second, or red star, shall have a visual spectrum centroid at 6563 * 10 millimicrons wavelength. The light emitted by the third, or blue star, shall have a visual spectrum centroid at 486.2 ± 5 miilimicrons wavelength. An auxiliary telescope shall be provided having a magnification of not less than 3 nor more than 6 diameters.

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4.5.4.4.2 Test procedure. In all focusing and testing, the particular aperture being observed should be centered in the fieid of view of the telescope under test and the image observed with the aid of the auxiliary telescope Focus the telescope under test on the white star for best image as seen through the auxiliary telescope The star image shall appear round with a sharply defined edge and shall be free from points, flares, rays, fog, haze, and halos other than the normal diffraction rings. No difference in coloration between one side of the star and the other shall be seen. Defocus the telescope under test slightly in either direction. The blurred image of the star shall remain round. Focus the telescope under test successively on the red star, adjusting for best focus as seen through the auxiliary teiescope and observe whether the blue star is also in focus

4.5.4.4.3 Resolution. The telescope shall be tested for resolution to determine conformance to 3.3.3. The target for the resolution test shall be of the Foucault resolving power pattern containing horizontal and verticai black bars on a white field. it shall be brightly illimunated by tungsten light without filters. Variation of target illumination to obtain opimum resolution is permitted. The auxiliary telescope should have a magnification of not less than 3 nor more than 6 diameters. The telescope under test shall be focused on the target for best image as seen through the auxiliary telescope.

4.5.4.4.4 Eypiece focus. The eyepiece shall be tasted for proper setting using a diopterscope of at least four power (see 6.4). Motion of the diopterscope eyepiece away from its objective is considered negative and toward it positive. Focus the diopterscope on a distant target and read the scale introduce diffused light into the objective of the telescope under test. With the objective of the diopterscope at the exit pupil of the telescope under test, focus the diopterscioe for clearest image of the reticle of the telescope and again read the scale. The eyepiece setting is calculated as the reciprocal of this distance in meters.

4.5.5 Environmental tests

4.5.5.1 Shock. The telescope shall be subjected to shock tests in accordance with MIL-S-901 for the grade, ciassification, and class of equipment specified therein. Prior to shock tests, the position of controls and adjustments shall be recorded. The reticle lamp shall be illuminated during shock tests. After being subjected to shock test, the telescope shall be deemed acceptable if it shows no damage, controls and adjustments have not appreciably shifted, and performance as specified herein is not impaired.

4.5.5.2 Vibration. The telescope shall be subjected to Type I environmental vibration tests specified in MiL-STD-167. Prior to vibration, the position of controls and adjustments shall be recorded. The reticle lamp shall be illuminated during vibration tests. Upon completion of vibration tests the telescope shall be deemed acceptable if it shows no damage, controls and adjustments have not appreciably shifted, and performance as specified herein is not impaired.

4.5.5.3 Temperate and humidity. The telescope shall be subjected to the temperature and humidity tests specified in MiL-F-18870 for Class 2 equipment.

4.5.5.4 Enclosure. The telescope shall be charged with dry air to a pressure of 4 psi. Drying and charging procedures should be in accordance with OP 3750. The exact internal air pressure, ambient temperature, and barometric pressure shall be simultaneously recorded. The air inlet valve shall then be closed to seal the telescope and the time recorded. The telescope shall then be subjected to the following temperatures in the order listed:

a. Room temperature (approximately 72 °F) until stabilized
b. Ambient temperature of +155 °F for at least 5 hours
c Room temperature (appoximateiy 72 °F) for at least 2 hours
d. Ambient temperature of -20 °F for at least 16 hours
e. Room temperature (approximately 72 °F) for approximanately 24 hours

During and at the end of the -20 °F temperature run of d. above the telescope shall be examined to determine that no condensation or moisture has collected on the internal surfaces of the telescope and on the surfaces of optical components At the end of the temperature run of e. above, the telescope internal air pressure ambient

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temperature and barometric pressure shall be simutaneously recoded and the internal air pressure corrected for any change between the ambient temperature and barometric pressure at the end of the test and the ambient temperature pressure at the beginning of the test and at intervals of not less than 8 hours for the duration thereof, each external knob and and energent shaft shall be operated at least one turn in each direction or if restricted to less than one turn, to the limit in each direction for at least one complete cycle.

4.5.5.4.1 Corrected and interpolated internal pressure. A check shall be made to determine that the corrected final internal air pressure is not less than the interpolated internal air pressure computed on the basis of 1 psi loss in 72 hours The interpolated air pressure shall be computed from the formula.

$$P_t = P_o e^{-0.004t}$$

Where:

P_t=interpolated pressure in psi at time t
P_o= Initial pressure in psi
e= Base of natural logarithrns
t = Times in hours between initial and final readings of internal air pressure in psi.

4.5.5.5 Static loading. The telescope shall be subjected to a static load test to determine that the instrument will withstand static loading of 1,000 pounds per square inch in all directions.

5. PACKAGING

5.1 Preservation.

5.1.1 Level A.

5.1.1.1 Cleaning, drying, and preservative application. Not applicblele.

5.1.1.2 Unit packaging. Unit packaging of the telescope shall be in accordance wiht Method 11a of Specification MIL-P-116. The container shall be in accordance with the drawings listed on LD258406.

5.1.2 Level C.

5.1.2.1 Cleaning, drying, and preservative application. Not applicable.

5.1.2.2 Unit packaging. Unit packaging of the telescope shall be in accordance with Method III of Specification MiL-P-116. The container shall be in accordance with the drawings listed on LD258406.

5.2 Packing. No overpacking is required for Level A or Level C

5.3 Marking.

5.3.1 Special markings. None, unless otherwise specified in the contractor order (see 6.2.1).

5.3.2 Normal markings. in addition to the markings required by contract or waler., unit packages and shipping containers shall be marked in accordance with the requirements of MIL-D-129, including serial numbers.

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6. NOTES

6.1 Intended use. Telescope Mk 100 Mod 0 or Mod 1, when installed in Gun Director Mk 68, is intended for use as a sighting and tracking instrument.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title number, and date of this specification
b. If first article sample is not required (see 3.1)
c. Quantity of telescopes required for the first article sample (see 3.1 and 4.4.1)
d. Testing activity for first article sample (see 4.4 and 6.3)
e. Sampiing plan and testing activity for quality conformance inspection (see 4.5.1)
f. Dimensions of reticle pattern required (see 3.3.4)
g. Additional tests and examinations required, if any, for first article inspection (see 4.4)
h. Levels required in preservation (see section 5)
i. Government furnished equipment to be supplied (see 3.2.1.3, 3.3.4.1, and 3.5)
j. Government loaned equipment to be furnished (see 3.6).

6.3 First article. When a first article inspection is required, the item(s) should be a first article sample. The first article should consist of one unit. The contracting offier should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and deposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definition.

6.4.1 Diopterscope. A dioptersope is an auxiliary telescope with a calibrated focusing motion for measuring the convergence or divergence of light penciis. The scale records directly the reciprocal distance to the focus of the pencil in diopters, the reference point on the diopterscope usually being the front of the objective

6.5 Subject term (key word) listing.

Diopterscope
Eyepiece
Gun director
Lenses
Magnifcation
Mirror
Ocular
Reticle
Telescope

6.6 Changes from previous issue Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity
Navy - 0S
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-81333A(OS)	TELESCOPE, MARK 100 MOD 0 AND 1

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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